

CN Unit 2 — Data Link Layer & MAC Complete Notes

Computer Networks (CN) · Unit 2 · Data Link Layer & MAC · 2024

Overview

These notes cover **Data Link Layer & MAC**. From bits on the wire to HTTP — layered network architectures, routing, transport protocols and the applications on top. Each section states the core idea, a worked intuition and the points examiners look for.

1. Framing

Character count, byte stuffing and bit stuffing mark frame boundaries.

Why it matters: Framing appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of framing in one line.
- A small example showing framing in practice.
- Common mistakes students make and how to avoid them.

2. Error Detection

Parity, checksums and CRC detect transmission errors.

Why it matters: Error Detection appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of error detection in one line.
- A small example showing error detection in practice.
- Common mistakes students make and how to avoid them.

3. Hamming Code

Hamming codes add parity bits that can locate and correct single-bit errors.

Why it matters: Hamming Code appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of hamming code in one line.
- A small example showing hamming code in practice.
- Common mistakes students make and how to avoid them.

4. Sliding Window Protocols

Stop-and-wait, Go-Back-N and Selective Repeat provide reliable, flow-controlled delivery.

Why it matters: Sliding Window Protocols appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of sliding window protocols in one line.
- A small example showing sliding window protocols in practice.
- Common mistakes students make and how to avoid them.

5. ALOHA and CSMA/CD

Pure and slotted ALOHA and CSMA/CD govern access to a shared medium.

Why it matters: ALOHA and CSMA/CD appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of aloha and csma/cd in one line.
- A small example showing aloha and csma/cd in practice.
- Common mistakes students make and how to avoid them.

6. Ethernet and Switching

Ethernet frames carry MAC addresses; learning switches build forwarding tables.

Why it matters: Ethernet and Switching appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of ethernet and switching in one line.
- A small example showing ethernet and switching in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Framing · Error Detection · Hamming Code · Sliding Window Protocols · ALOHA and CSMA/CD · Ethernet and Switching